

Arrays in JS

Collections of items

c- `int arr[5]={10,20,30}`

js- `let arr=[10,20,30];`

c- `char name[6]={'p','r','i','n','c','e'}` (array of chars is a string in c)

js- `let arr=["rahul","Delhi"];`(here strings can be inputted directly)

Create Array

`let heroes = [ "ironman", "hulk", "thor", "batman" ];`

`let marks = [ 96, 75, 48, 83, 66 ];`

`let info = [ "rahul", 86, "Delhi" ];`

(valid)but not preferred.

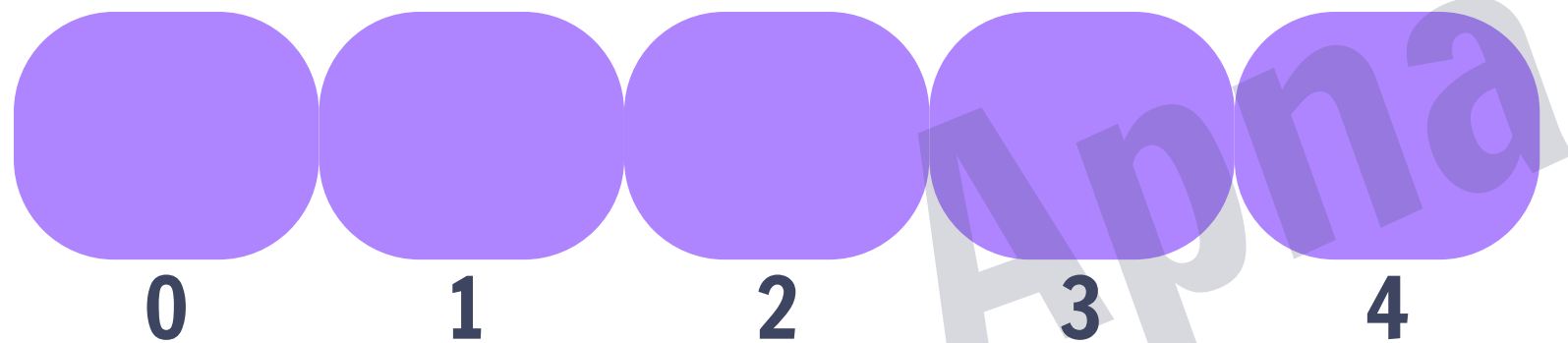
arrays are used instead of objects coz in objects u need to give new name to every key but here only 1 name and all the data is here bu just 1 variable call (name of the array)

not necessarily similar datatypes like in c you can make a array of diff datatypes too like here , let info has, but we dont generally prefer doing that and only make arrays for similar datatypes but if u wanna for diff datatypes , u can in js.

Arrays in JS

Array Indices

arr[0], arr[1], arr[2]



interesting the datatype of a array in js is a 'object';

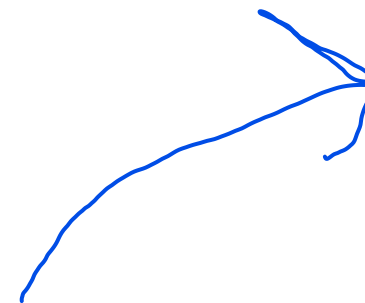
but in c theres no such thing int arr[5] ka datatype is "arr of 5 ints" datatype of char name[20]; is "array of 20 chars"

array is a object but instead of key:values it has index:value (arr[0]:3) like this .

coz strings-immutable
arrays-mutable

you can also change the indices values j like c
arr[0]=66;

but u cant do that for strings as uk it has strcpy



Looping over an Array

Print all elements of an array

iterable are those items in which we can iterate.(use i or loops) like strings arrays objects

these 3 are collection of items

```
for(let i=0;i<arr.length;i++){
```

```
  console.log(arr[i]);  
}
```

where n is the length of array.,
in c --> n=sizeof(arr)/sizeof(arr[0]);
in js --> n=arr.length;

OR

```
for(let i=1;i<=arr.length;i++){
```

```
  console.log(arr[i-1]);  
}
```

Can also use for-of or for-in loops... generally for arrays we use for-of ...

check code for this for-of loop in action.....

Let's Practice

**Qs. For a given array with marks of students -> [85, 97, 44, 37, 76, 60]
Find the average marks of the entire class.**

Apna College

Let's Practice

Qs. For a given array with prices of 5 items -> [250, 645, 300, 900, 50]

All items have an offer of 10% OFF on them. Change the array to store final price after applying offer.

since strings are immutable their methods always returned a new str and didnt actually changed the string ,
but here since array is mutable some methods change the array directly and some return a copy of it.

Arrays in JS

Array Methods

`arr.push("new element");`

the methods name are written like Push()
but they're push();

Push() : add to end (changes the array directly) line 91-95 check em.

Pop() : delete from end & return (changes the array directly) doesn't print the array by itself u need to do console.log, see line 98-102

deletes from end and returns that deleted value which won't be shown until u assign it to a variable and then print that variable.... like `let deleted=food.pop(); console.log("deleted",deleted);`

toString() : converts array to string (doesn't change the array) just returns a new str
(also returns that string)

makes the array a string with commas like new string is 98,96,101
or it is banana,apple,grapes like this not like bananaapplegrapes...
NO. see line 104- 110

Arrays in JS

Array Methods

`arr1.concat(arr2);`



Concat() : joins multiple arrays & returns result (doesn't change the array originally)

Unshift() : add to start (changes the array directly)

j like push() method

shift() : delete from start & return (changes the array directly)

j like pop() method

Arrays in JS

Array Methods

part of array gets returned j like in strings...

Slice() : returns a piece of the array

(doesn't change the array directly)

`slice( startIdx, endIdx )`

write the start and end index but the end index doesn't count only till start and one less than what u enter as end see code line 144.

Splice() : change original array (add, remove, replace)

(made for changing the original array so ofc does it)

`splice( startIdx, delCount, newEl1... )`

these are what splice is used for, all 3 are shown in code
162(add), 167(remove), 158(replace)

(this is the real syntax and values to be added in it)

starting index is compulsory then whether to remove or without removing just adding new element depends so remove is optional and even replace is optional if u just wanna remove smth and don't wanna replace it

doesn't return anything so printing the statement `arr.splice();` will not give anything just do `splice();` and then print the original array;

Let's Practice

Qs. Create an array to store companies -> “Bloomberg”, “Microsoft”, “Uber”, “Google”, “IBM”, “Netflix”

a. Remove the first company from the array

b. Remove Uber & Add Ola in its place

c. Add Amazon at the end